Healthcare Provider Fraud Risk Detection Using Claims Analytics, Risk Scoring, and Machine Learning

Abstract

Healthcare provider fraud can contribute to unnecessary healthcare spending and requires effective methods for identifying unusual billing and utilization behavior. This project developed an end-to-end fraud-risk detection framework using provider, claims, and beneficiary data. The workflow included SQL-based data preparation, Python data cleaning, provider-level feature engineering, exploratory data analysis, rule-based risk scoring, and machine-learning classification.

The training dataset contained 5,410 providers, including 506 labeled as Potential Fraud. Seven provider characteristics were used to create an interpretable risk score based on 90th-percentile thresholds, while 33 engineered features were used to train Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting models. Gradient Boosting was selected as the final model, achieving 0.90 fraud recall, 0.65 F1-score, 0.9674 ROC-AUC, and 0.7825 PR-AUC on the validation set. Feature-importance analysis identified total reimbursement, maximum claim duration, and claims per patient as the most influential predictors.

When applied to the separate unlabeled test dataset, the model classified 227 of 1,353 providers as Potential Fraud. The results demonstrate that combining interpretable risk scoring with machine learning can support healthcare fraud screening and help prioritize providers for further investigation, while model outputs should be treated as indicators of risk rather than definitive evidence of fraud.

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1. Introduction

Healthcare fraud is a significant challenge for healthcare systems because fraudulent or abusive billing practices can increase healthcare costs, waste limited financial resources, and make it more difficult for insurers and public programs to identify where spending is legitimate. Suspicious activity may appear through unusually high claim volumes, excessive reimbursements, repeated or high-cost claims, abnormal utilization patterns, or provider behavior that differs substantially from that of comparable providers. However, these patterns do not automatically indicate fraud. A provider may legitimately have high reimbursement or utilization because of patient complexity, service type, or practice size. For this reason, healthcare fraud detection is better approached as a risk-screening problem, where data analytics are used to identify unusual provider behavior that may warrant additional investigation rather than to establish fraud directly.

This project developed an end-to-end healthcare provider fraud-risk detection workflow using provider, claims, and beneficiary data. The analysis began by organizing and examining the raw healthcare data (Gupta, n.d.) and establishing the relationships between providers, individual claims, and beneficiaries through the Provider, ClaimID, and BeneID identifiers. Claims and beneficiary information were then integrated so that provider billing behavior could be studied alongside the demographic and clinical characteristics of the patients treated by each provider. The data were cleaned and validated before being transformed into a provider-level analytical dataset containing billing, utilization, demographic, chronic-condition, and additional fraud-risk features.

The project then used feature engineering and exploratory data analysis to examine differences between providers labeled as potentially fraudulent and those not labeled as potentially fraudulent. Provider-level characteristics included claim volume, reimbursement patterns, claims per patient, inpatient utilization, patient age and demographic composition, chronic-disease burden, claim duration, and high-cost claim activity. An interpretable rule-based risk-scoring framework was also developed using selected provider characteristics and 90th-percentile training thresholds. Providers were assigned risk indicators and classified into Low, Medium, or High risk categories according to their combined risk score.

Finally, the provider-level dataset was used to train and compare several machine-learning classification models, including Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting. The training data were divided using a stratified 80/20 split, and model performance was assessed using

precision, recall, F1-score, confusion matrices, ROC-AUC, and PR-AUC. The selected model was then applied to a separate unlabeled test-provider dataset to generate predicted fraud labels and fraud probabilities. Together, these methods provide both an interpretable view of suspicious provider behavior and a predictive framework for prioritizing providers that may require further review.

1.1 Project Objective

The objective of this project is to identify healthcare providers with elevated potential fraud risk by analyzing provider billing behavior, healthcare utilization, beneficiary characteristics, chronic-condition patterns, and engineered claim-risk indicators.

The project develops both an interpretable rule-based risk scoring framework and a machine-learning classification framework. These approaches are used together to examine suspicious provider behavior, compare fraud-labeled and non-fraud-labeled providers, and estimate fraud probability for providers in an unlabeled test dataset.

1.2 Research Questions

1. What demographic and chronic-condition patterns characterize the patient populations treated by different healthcare providers?
2. Do providers labeled as potentially fraudulent demonstrate different claim volume, reimbursement, utilization, or billing patterns compared with providers not labeled as potentially fraudulent?
3. Which provider-level features appear most useful for distinguishing elevated fraud risk?
4. Can an interpretable rule-based scoring system classify providers into Low, Medium, and High fraud-risk categories?
5. Can machine-learning models use provider-level claims, demographic, chronic-condition, and risk-related features to predict potential provider fraud?
6. When the selected model is applied to an unlabeled test dataset, which providers are identified as having elevated predicted fraud risk?

2. Dataset and Data Structure

The dataset used in this project was obtained from the Healthcare Provider Fraud Detection Analysis dataset available on Kaggle. The data are organized into separate provider, claims, and beneficiary files for training and test populations. This structure makes it possible to examine provider billing behavior while also incorporating information about individual claims and the demographic and chronic-condition characteristics of beneficiaries receiving care.

The training data contain information for 5,410 healthcare providers, with each provider assigned a Potential Fraud label indicating whether the provider was identified as potentially fraudulent or not potentially fraudulent. Of these providers, 506 were labeled Potential Fraud and 4,904 were labeled Not Potential Fraud. The training claims consist of 40,474 inpatient claims and 517,737 outpatient claims, producing 558,211 combined claims after inpatient and outpatient records were consolidated. These claims are associated with 138,556 beneficiaries.

A separate unlabeled test dataset was used for final fraud-risk prediction. The test data contain 1,353 healthcare providers, 9,551 inpatient claims, 125,841 outpatient claims, and 135,392 combined claims involving 63,968 beneficiaries. Unlike the training provider data, the test provider file does not contain a known fraud label. Therefore, it was used only after model development to generate predicted fraud classifications and probabilities rather than to evaluate model accuracy.

2.1 Dataset Components

The project used three primary types of healthcare data: provider-level data, claim-level data, and beneficiary-level data. Provider data summarized billing and utilization behavior and, for the training set, included the PotentialFraud label. Claim-level data contained inpatient and outpatient service information such as claim dates, reimbursement amounts, deductibles, physician information, diagnosis codes, and procedure codes. Beneficiary data provided patient-level demographic characteristics, chronic-condition indicators, renal disease status, and related healthcare information. These datasets were linked using the Provider, ClaimID, and BeneID identifiers to create the analytical dataset used throughout the project.

Table 1. Summary of Training and Test Data

Data	Training	Test
Providers	5,410	1,353
Beneficiaries	138,556	63,968
Inpatient claims	40,474	9,551
Outpatient claims	517,737	125,841
Combined claims	558,211	135,392
Potential Fraud providers	506	Not available
Not Potential Fraud providers	4,904	Not available

2.2 Data Relationships

The three main components of the dataset are connected through a set of common identifiers. Each healthcare provider is represented by a unique Provider identifier. Individual healthcare claims are identified through ClaimID, while beneficiaries are identified using BeneID. A claim therefore links a beneficiary to the provider responsible for submitting that claim.

The analytical relationship can be summarized as:
Provider → Claims → Beneficiary

A provider may submit multiple claims, and the same beneficiary may appear in multiple claims. Beneficiary information was therefore merged with claim-level data using BeneID, while claim information was aggregated by Provider to create one analytical record for each healthcare provider. This provider-level structure was subsequently used for exploratory analysis, risk scoring, and machine-learning model development.

2.3 Target Variable

The machine-learning models were trained using PotentialFraud as the outcome variable. In the training data, providers were classified as either Potential Fraud or Not Potential Fraud. For modeling purposes, this categorical label was converted into a binary variable called PotentialFraudFlag, where a value of 1 represented Potential Fraud and 0 represented Not Potential Fraud.

The separate test dataset did not contain known fraud labels. Therefore, it was used only after model training to generate predicted fraud classifications and fraud probabilities for the 1,353 test providers. These predictions indicate estimated fraud risk and should not be interpreted as confirmed evidence of fraud.

3. Data Preparation and SQL Processing

The initial stage of the project focused on organizing the raw healthcare data and creating a consistent provider-level structure for analysis. SQL was used to inspect the training and test tables, review row counts and column structures, verify provider and claim identifiers, and examine the distribution of the PotentialFraud label in the training data. Separate inpatient, outpatient, beneficiary, and provider tables were used as the starting point for this stage.

The inpatient and outpatient claim tables were linked with beneficiary information using BeneID. For the training data, claim records were also linked with the provider fraud label using the Provider identifier. Separate joined inpatient and outpatient views were created and then combined into a single claim-level dataset by adding a claim_type indicator to distinguish inpatient from outpatient services. This produced a unified structure that could be used consistently for provider-level aggregation.

The combined claim-level data were then aggregated by the provider to summarize healthcare utilization and billing behavior. Initial provider-level features included total claims, unique claims, unique patients, inpatient and outpatient claim counts, total and average reimbursement, total and average deductible amounts, average claim duration, and claims per patient. Additional utilization measures were calculated, including inpatient and outpatient claim ratios, reimbursement per patient, and deductible per patient. The resulting provider-level table formed the base analytical dataset that was later enriched in Python with demographic, chronic-condition, and additional fraud-risk features.

The same SQL preparation process was reproduced for the unlabeled test data to maintain consistency between training and prediction datasets. Because the test provider data did not include a fraud label, the test SQL pipeline generated the same core billing and utilization variables without PotentialFraud. This ensured that the test providers could later undergo the same Python-based feature-engineering process and be aligned with the features used during model development.

4. Data Cleaning in Python

Following the SQL-based data preparation, the training datasets were cleaned and validated in Python before feature engineering. Separate cleaning procedures were applied to the claims, beneficiary, and provider datasets to standardize data types, address missing values, recode categorical indicators, and identify potentially invalid values. The cleaned datasets were then merged and checked to ensure that the analytical records remained complete.

4.1 Claims Data Cleaning

The claim start and end date variables were converted from text to datetime format to support claim-duration calculations and chronological validation. No missing values were identified in either ClaimStartDt or ClaimEndDt after conversion. A date-consistency check also confirmed that there were no claims with an end date occurring before the corresponding start date.

Missing values were identified in DeductibleAmtPaid for 899 claims, all of which were inpatient claims. A binary missingness indicator was retained to record which claims originally contained a missing deductible value. The missing deductible amounts were then imputed using the median inpatient deductible of 1,068, resulting in no remaining missing deductible values. Additional claim-level validation found no negative reimbursement amounts and no negative deductible amounts. These checks supported the use of the cleaned claims data for subsequent provider-level aggregation and feature engineering.

4.2 Beneficiary Data Cleaning

Beneficiary date-of-birth (DOB) and date-of-death (DOD) variables were converted to datetime format. A chronological validation confirmed that no recorded death date occurred before the corresponding date of birth. Recorded dates of birth ranged from 1909 to 1983.

The beneficiary dataset contained several health-condition indicators that were standardized for numerical analysis. RenalDiseaseIndicator was converted from Y and 0 into a binary format, where 1 indicated renal disease and 0 indicated its absence; after encoding, 19,578 beneficiaries were identified with renal disease and 118,978 without it. The chronic-condition variables were also converted into binary indicators by

recoding the original value 2, which represented absence of a condition, to 0, while retaining 1 for condition presence. This standardization was applied to Alzheimer's disease, heart failure, kidney disease, cancer, obstructive pulmonary disease, depression, diabetes, ischemic heart disease, osteoporosis, rheumatoid arthritis, and stroke, allowing the conditions to be used consistently in later prevalence and chronic-disease burden calculations.

Demographic variables including gender, race, state, and county were also inspected for missing or unexpected values. No missing values were found in race, state, or county in the inspected training data. Coverage-month variables for Medicare Part A and Part B were retained in their existing numerical form after inspection.

Beneficiary-level annual reimbursement variables were also reviewed for unusual values. The analysis identified 15 negative inpatient annual reimbursement values and 12 negative outpatient annual reimbursement values. These observations were retained rather than automatically removed because they occurred in the source beneficiary data and were treated as recorded financial adjustments rather than claim-level data-entry errors. Importantly, the individual claim-level reimbursement data used in provider billing calculations contained no negative values. The cleaning script explicitly inspected these annual reimbursement fields rather than silently altering them.

4.3 Provider Data Cleaning

The training provider dataset contained 5,410 provider-level records. The categorical PotentialFraud field was converted into a binary variable, PotentialFraudFlag, with 1 representing Potential Fraud and 0 representing Not Potential Fraud. The dataset included 506 providers labeled as Potential Fraud and 4,904 labeled as Not Potential Fraud, and no missing values were introduced during the encoding process. The provider-level billing and utilization variables were also checked for invalid values. No negative values were found in the main claim, reimbursement, deductible, claims-per-patient, or reimbursement-per-patient measures, and both inpatient and outpatient claim ratios were confirmed to fall within the expected range of 0 to 1.

4.4 Claims and Beneficiary Data Integration

After cleaning, the claims and beneficiary datasets were merged using BeneID to combine claim-level information with beneficiary demographic and clinical characteristics. The merge preserved all 558,211 claim records, and no missing values were found in key beneficiary fields such as DOB, Gender, and Race. The resulting merged dataset was then used for provider-level demographic, chronic-condition, and fraud-risk feature engineering.

5. Feature Engineering

Following data cleaning and integration, additional features were created to represent the demographic characteristics, chronic-condition burden, and claim behavior associated with each provider. Because the final analysis and machine-learning models operate at the provider level, claim- and beneficiary-level information was aggregated so that each provider was represented by a single analytical record.

5.1 Demographic Features

Patient age was calculated at the time of each claim using the beneficiary date of birth and claim start date. Across all claims, patient age ranged from 25 to 100 years, with a mean age of approximately 72.8 years. To avoid counting the same beneficiary multiple times for the same provider, a provider-beneficiary table was created containing one record for each unique provider-patient combination. This resulted in 363,300 unique provider-beneficiary relationships across all 5,410 providers.

Provider-level demographic features were then calculated, including the number of unique patients, average and median patient age, gender composition, and race distribution. Gender and race counts were converted into percentages so that providers with different patient volumes could be compared on a common scale. The final demographic feature table contained 18 variables for each of the 5,410 providers.

5.2 Chronic-Condition Features

Chronic-condition information was aggregated at the provider level to characterize the health profile of patients treated by each provider. For each unique provider-patient relationship, the total number of recorded chronic conditions was calculated across the available disease indicators. The chronic-condition count ranged from 0 to 11, with an average of approximately 4.32 conditions per patient.

Provider-level features included the average and median chronic-condition count, as well as disease-specific patient counts and prevalence rates for conditions such as Alzheimer’s disease, heart failure, kidney disease, cancer, obstructive pulmonary disease, depression, diabetes, ischemic heart

disease, osteoporosis, rheumatoid arthritis, stroke, and renal disease. The resulting chronic-condition feature table contained 27 variables for all 5,410 providers, with no missing disease-count values.

5.3 Claim Duration Features

Claim duration was calculated as the number of days between the claim start and end dates. Across the training claims, duration ranged from 0 to 36 days, with an average of approximately 1.73 days. The median duration was zero because a large proportion of claims began and ended on the same date.

To summarize this behavior at provider level, two features were created: median claim duration and maximum claim duration. These variables were intended to capture typical claim length as well as unusually long claims associated with individual provide

5.4 High-Cost Claim Features

High-cost claims were identified separately for inpatient and outpatient services because reimbursement levels differ substantially between the two types of care. The calculated high-cost thresholds were $30,000 for inpatient claims and $1,500 for outpatient claims. Claims meeting or exceeding the relevant threshold were assigned a high-cost claim indicator.

A total of 29,198 claims were classified as high cost. This represented approximately 5.26% of inpatient claims and 5.23% of outpatient claims. These claim-level indicators were then aggregated to create two provider-level variables: the number of high-cost claims and the percentage of a provider's claims classified as high cost.

5.5 Final Provider Feature Structure

The demographic, chronic-condition, claim-duration, and high-cost claim features were subsequently combined with the existing provider-level billing and utilization measures. This expanded the analytical dataset beyond basic reimbursement and claim-volume information by incorporating characteristics of the patient population and additional indicators of provider behavior. All feature tables retained the same 5,410 providers, allowing them to be merged into the final training dataset used for exploratory analysis, risk scoring, and machine-learning modeling.

6. Exploratory Analysis

Exploratory data analysis was performed on the final provider-level training dataset to examine how providers labeled as Potential Fraud differed from those labeled as Not Potential Fraud. The analysis focused on fraud-label distribution, billing and utilization behavior, beneficiary demographics, chronic-condition profiles, claim-duration characteristics, and high-cost claim activity.

6.1 Fraud Label Distribution

The training dataset contained 5,410 providers, of which 506 (9.35%) were labeled Potential Fraud and 4,904 (90.65%) were labeled Not Potential Fraud. This substantial class imbalance was considered later during model development and evaluation

6.2 Billing and Utilization Patterns

The largest differences between the two provider groups were observed in billing volume and reimbursement. Potential Fraud providers averaged approximately 420.55 claims, compared with 70.44 claims among Not Potential Fraud providers. Their average total reimbursement was also substantially higher, at approximately $584,350 compared with $53,194. Reimbursement per patient averaged about $4,557 for Potential Fraud providers versus $1,705 for the comparison group. Potential Fraud providers also had a higher average inpatient claim ratio (0.33 vs. 0.13) and higher average deductible amounts. These findings suggest that provider claim volume, reimbursement intensity, and inpatient utilization are important characteristics associated with the Potential Fraud label.

6.3 Demographic and Chronic-Condition Patterns

Patient demographic characteristics were generally similar between the two groups. Average patient age was approximately 72.7 years for both Potential Fraud and Not Potential Fraud providers, while gender and race distributions also showed only small differences. The major demographic distinction was patient volume: Potential Fraud providers treated an average of approximately 242 unique patients, compared with 49 unique patients for Not Potential Fraud providers.

Chronic-condition prevalence was modestly higher among patients treated by Potential Fraud providers. Average chronic-condition burden was 4.61 conditions compared with 4.41, with higher rates observed for several conditions, including heart failure, kidney disease, obstructive pulmonary disease, depression, diabetes, ischemic heart disease, and renal disease. However, these differences were considerably smaller than the differences observed in provider billing and utilization measures.

6.4 Claim-Duration and High-Cost Claim Patterns

Potential Fraud providers also demonstrated differences in claim-duration and high-cost claim activity. Their average maximum claim duration was approximately 26.97 days, compared with 15.07 days for Not Potential Fraud providers. They averaged approximately 22.06 high-cost claims, compared with 3.68 among Not Potential Fraud providers. However, the percentage of claims classified as high cost differed more modestly, at 5.87% versus 5.25%.

This indicates that the larger number of high-cost claims among Potential Fraud providers is partly associated with their substantially greater overall claim volume, while the proportion of high-cost claims shows a smaller difference.

6.5 Correlation and Key Findings

Correlation analysis showed that total reimbursement had the strongest observed positive correlation with the fraud label ($r = 0.58$), followed by total deductible and inpatient claim volume ($r = 0.53$ for both). Several disease-patient counts also showed moderate positive correlations with the fraud label.

The analysis also identified substantial correlation among several provider-volume variables. For example, total claims and unique claims were perfectly correlated, while unique-patient counts were strongly related to gender, race, and disease-patient counts. This suggests that many raw count variables partly reflect provider size rather than independent fraud characteristics. For this reason, redundant variables were considered during later machine-learning feature preparation, while normalized measures such as reimbursement per patient, claims per patient, disease rates, and claim ratios provided additional context.

Table 2. Selected EDA Comparison by Fraud Label

Feature	Not Potential Fraud	Potential Fraud
Average total claims	70.44	420.55
Average unique patients	49.11	242.02
Average total reimbursement	$53,193.72	$584,350.04
Average reimbursement per patient	$1,705.39	$4,557.47
Average inpatient claim ratio	0.13	0.33
Average patient age	72.70	72.71
Average chronic-condition count	4.41	4.61
Average maximum claim duration	15.07 days	26.97 days
Average high-cost claim count	3.68	22.06
Average high-cost claim percentage	5.25%	5.87%

Fig 1. Provider Fraud Label Distribution



Fig 2. Average Total Claims by Fraud Label

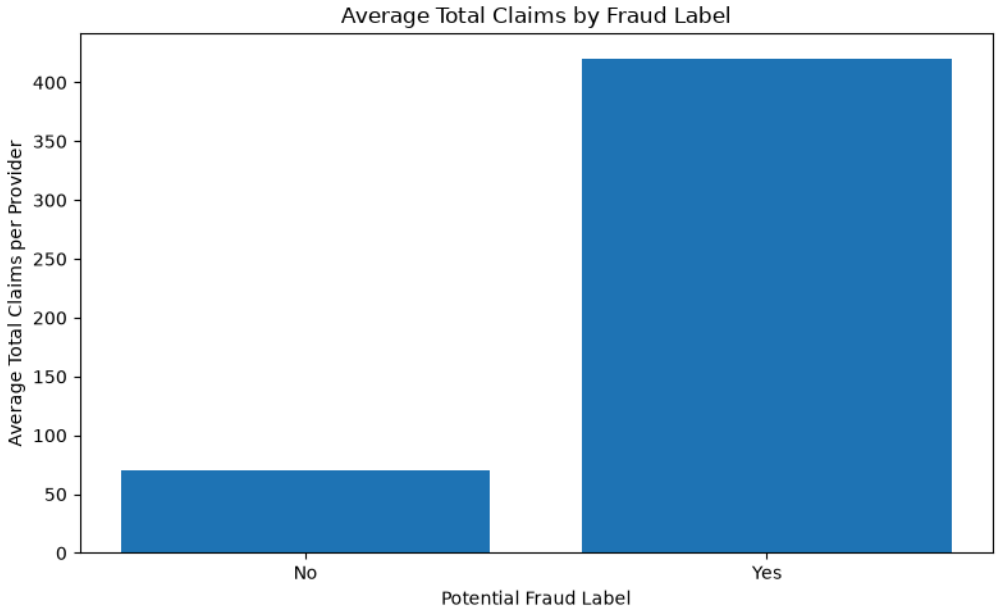


Fig 3. Average Total Reimbursement by Fraud Label

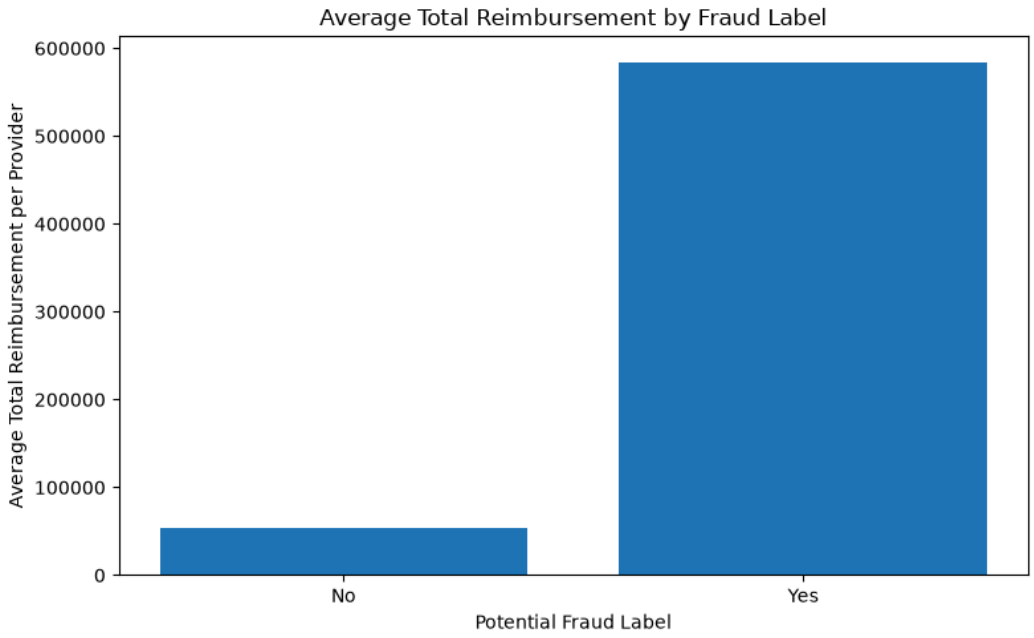
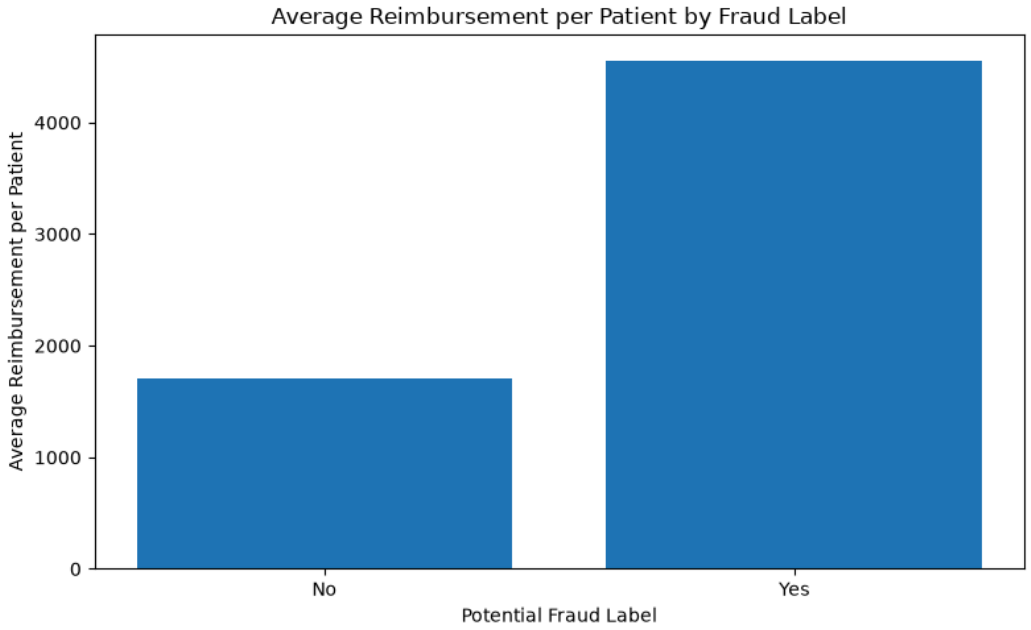


Fig 4. Average Reimbursement per Patient or Inpatient Claim Ratio by Fraud Label



6.6 Key EDA Findings

The exploratory analysis showed that Potential Fraud providers were most clearly associated with higher claim volumes, substantially greater reimbursement, increased inpatient utilization, longer maximum claim durations, and more high-cost claims. These patterns suggest that billing intensity and service utilization were the strongest signals of elevated fraud risk in the training data.

By comparison, patient age, gender, and race distributions were very similar across the two fraud groups, while chronic-condition differences were relatively modest. This indicates that provider behavior and billing patterns were more informative for fraud-risk identification than basic demographic characteristics.

7. Rule-Based Provider Risk Scoring

To provide an interpretable measure of provider fraud risk, a rule-based scoring system was developed using seven billing and utilization features. The 90th percentile of the training data was used as the high-risk threshold for each feature. Providers received one point for every threshold they met or exceeded. The selected indicators were total claims, total reimbursement, claims per patient, reimbursement per patient, inpatient claim ratio, high-cost claim percentage, and maximum claim duration. These points were summed to create a total risk score from 0 to 7.

7.1 Risk Thresholds and Risk Levels

The rule-based risk score used the 90th percentile of the training data as the threshold for seven provider characteristics: total claims (228), total reimbursement ($241,813), claims per patient (2.05), reimbursement per patient ($5,890.91), inpatient claim ratio (0.58), high-cost claim percentage (10.81%), and maximum claim duration (23 days). Providers received one risk point for each threshold they met or exceeded. Based on the total score, providers were classified as Low risk (0–1 points), Medium risk (2–3 points), or High risk (4 or more points). Among the 5,410 training providers, 4,407 were classified as Low risk, 867 as Medium risk, and 136 as High risk.

7.2 Risk Level and Fraud Label

The risk categories showed a clear relationship with the existing fraud labels. Among High-risk providers, 75.74% were labeled Potential Fraud, compared with 32.53% of Medium-risk providers and only 2.75% of Low-risk providers. This pattern indicates that the rule-based scoring system successfully concentrated a large proportion of fraud-labeled providers in the higher-risk categories while maintaining an interpretable approach to provider screening.

Table 3. Risk Level by Fraud Label

Risk Level	Not Potential Fraud	Potential Fraud	Fraud Percentage
Low	4,286	121	2.75%
Medium	585	282	32.53%
High	33	103	75.74%

8. Machine Learning Methodology

8.1 Model Data Preparation

The final provider-level training dataset contained 5,410 providers and 64 variables. PotentialFraudFlag was used as the target variable, while identifiers, target-related fields, redundant count variables, and highly correlated features were excluded from model input. This resulted in 33 predictor features covering billing, utilization, demographics, chronic-condition rates, claim duration, and high-cost claim behavior. No missing values were present in the selected modeling features.

8.2 Training and Validation Split

The labeled provider data were divided into 80% training and 20% validation sets using stratified sampling and a random state of 42. The training set contained 4,328 providers, including 3,923 Not Potential Fraud and 405 Potential Fraud providers. The validation set contained 1,082 providers, including 981 Not Potential Fraud and 101 Potential Fraud providers. Stratification preserved the original class distribution across both subsets.

Because Potential Fraud providers represented a relatively small proportion of the dataset, class imbalance was considered during model training. Balanced class weights were used for Logistic Regression, Decision Tree, and Random Forest, while balanced sample weights were applied during Gradient Boosting training.

8.3 Models Evaluated

Four supervised classification algorithms were compared: Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting. Logistic Regression provided a linear baseline, while Decision Tree and Random Forest captured nonlinear relationships between provider characteristics and fraud risk. Gradient Boosting was included as an ensemble method capable of learning more complex interactions among the engineered provider features.

8.4 Evaluation Metrics

Model performance was assessed using precision, recall, F1-score, confusion matrix, ROC-AUC, and PR-AUC. Particular attention was given to fraud-class recall because missing a potentially fraudulent provider is especially important in a screening setting. Precision and F1-score were also considered to evaluate the trade-off between detecting fraud-labeled providers and generating false-positive alerts. PR-AUC was particularly useful because of the imbalanced distribution of the target variable.

9. Model Results

The four models demonstrated generally strong discrimination, although their ability to identify the minority fraud class varied. Gradient Boosting and Random Forest produced the highest overall validation accuracy of approximately 91%, while Decision Tree produced the lowest at 88%.

Table 4. Validation Performance of Machine-Learning Models

Model	Precision	Recall	F1	ROC-AUC	PR-AUC
Logistic Regression	0.47	0.89	0.62	0.9669	**0.7892**
Decision Tree	0.43	0.87	0.58	0.8752	0.6626
Random Forest	**0.52**	0.88	**0.65**	0.9658	0.7635
Gradient Boosting	0.51	**0.90**	**0.65**	**0.9674**	0.7825

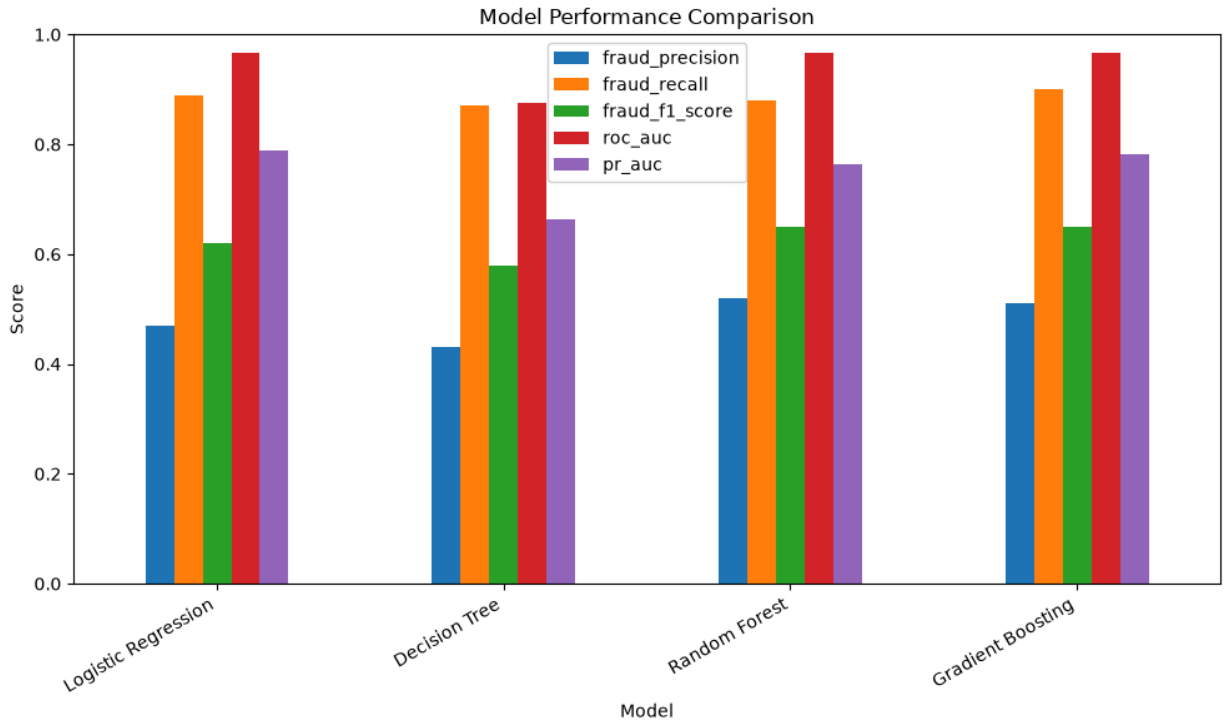
9.1 Model Comparison

Logistic Regression achieved a strong fraud recall of 0.89 and the highest PR-AUC of 0.7892, although its fraud precision was lower at 0.47. Decision Tree produced the weakest overall validation performance, with lower precision, F1-score, ROC-AUC, and PR-AUC.

Random Forest achieved the highest fraud precision (0.52) and an F1-score of 0.65, detecting 89 of the 101 fraud-labeled validation providers. However, its fraud recall of 0.88 was slightly lower than both Logistic Regression and Gradient Boosting.

Gradient Boosting provided the highest fraud recall (0.90) while maintaining an F1-score of 0.65, ROC-AUC of 0.9674, and PR-AUC of 0.7825. It correctly identified 91 of the 101 fraud-labeled providers in the validation set and missed 10.

Fig 5. Model Comparison



9.2 Selected Model

Gradient Boosting was selected as the final model because the project prioritized fraud detection recall while also considering F1-score, PR-AUC, and ROC-AUC. The model produced 91 true positives, 895 true negatives, 86 false positives, and 10 false negatives on the validation dataset.

Although Logistic Regression achieved a slightly higher PR-AUC and Random Forest slightly higher precision, Gradient Boosting provided the highest fraud recall while maintaining competitive performance across the remaining metrics. This made it the most appropriate choice for the project's purpose as a provider fraud-risk screening model, where reducing missed high-risk providers was considered particularly important. The model-ranking code likewise prioritized fraud recall first, followed by F1-score, PR-AUC, and ROC-AUC.

Table 5. Confusion matrix: Gradient Boosting

	Predicted Not Fraud	**Predicted Potential Fraud**
Actual Not Fraud	895	86
Actual Potential Fraud	10	91

9.3 ROC / PR Performance

ROC-AUC values were high for Logistic Regression (0.9669), Random Forest (0.9658), and Gradient Boosting (0.9674), indicating strong ability to distinguish between Potential Fraud and Not Potential Fraud providers. The Decision Tree performed considerably lower with a ROC-AUC of 0.8752.

PR-AUC values provided additional insight because the dataset was imbalanced. Logistic Regression achieved the highest PR-AUC (0.7892), followed closely by Gradient Boosting (0.7825), while Random Forest and Decision Tree achieved 0.7635 and 0.6626, respectively. Overall, Gradient Boosting maintained strong performance across both ROC-AUC and PR-AUC while also achieving the highest fraud recall.

10. Feature Importance

Feature importance was examined using the selected Gradient Boosting model to determine which provider characteristics contributed most strongly to fraud-risk prediction. The analysis showed that total reimbursement was by far the most influential feature, with an importance score of 0.7299. This was followed by maximum claim duration (0.1040) and claims per patient (0.0415). Other features, including total deductible, total claims, average claim duration, reimbursement per patient, demographic variables, and chronic-condition rates, contributed smaller amounts to the model.

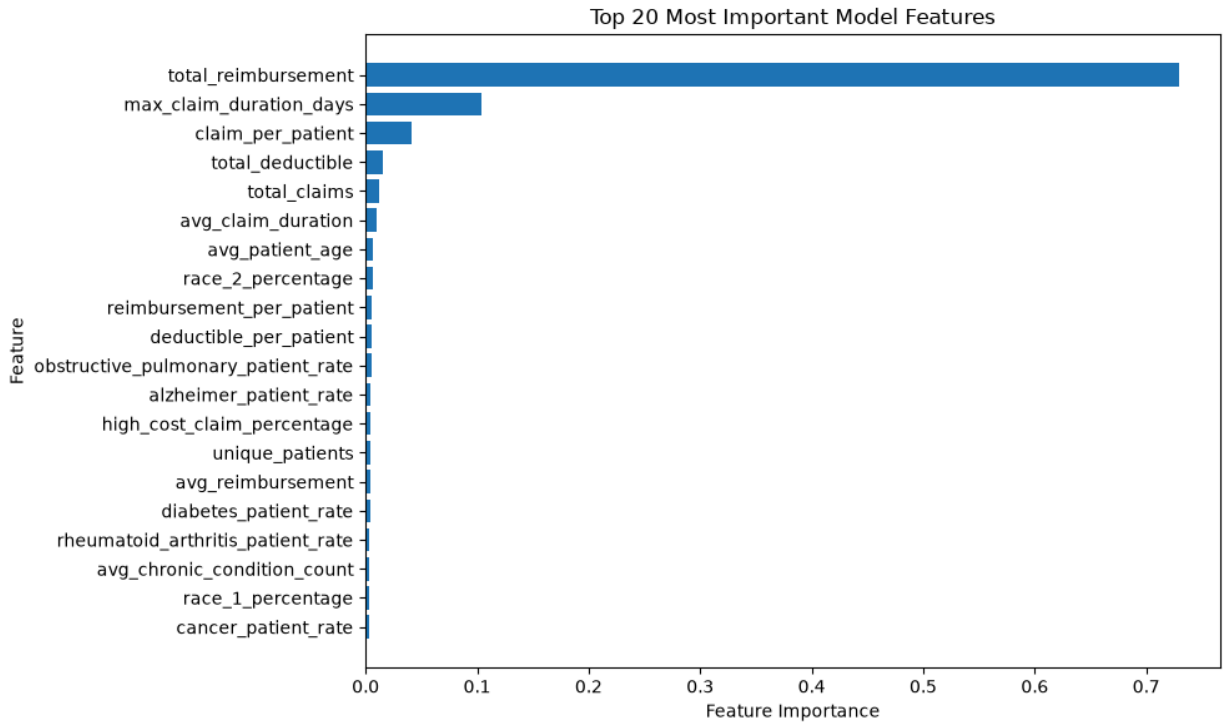
10.1 Most Important Predictors

The strong importance of total reimbursement is consistent with the exploratory analysis, where Potential Fraud providers showed substantially higher reimbursement levels. Maximum claim duration and claims per patient also contributed meaningfully, suggesting that both unusually long claims and greater service utilization per beneficiary helped distinguish higher-risk providers. Demographic and chronic-condition variables had comparatively lower individual importance.

10.2 Interpretation of Feature Importance

Overall, the feature-importance results indicate that the Gradient Boosting model relied primarily on billing and utilization characteristics rather than patient demographic characteristics. However, feature importance represents the relative contribution of variables to model predictions and does not imply that a feature directly causes fraudulent behavior.

Figure 6. Top 20 Most Important Features in the Gradient Boosting Model



11. Test Dataset Predictions

11.1 Application to the Test Dataset

The final test dataset contained 1,353 providers and 62 variables. After applying the same feature-selection process used during model training, 33 predictor variables were supplied to the selected Gradient Boosting model. The model was then used to generate a predicted fraud label and fraud probability for each provider.

11.2 Predicted Fraud Results

The Gradient Boosting model classified 227 providers as Potential Fraud and 1,126 providers as Not Potential Fraud. Predicted fraud probabilities ranged from approximately 0.006 to 0.989, with an average probability of 0.188. The highest predicted probability was observed for provider PRV52338 (0.9895), followed by PRV53105 (0.9877) and PRV54751 (0.9873).

Because the test dataset does not contain known fraud labels, these predictions represent estimated fraud risk rather than confirmed fraud outcomes.

Fig 7. Predicted Fraud Status

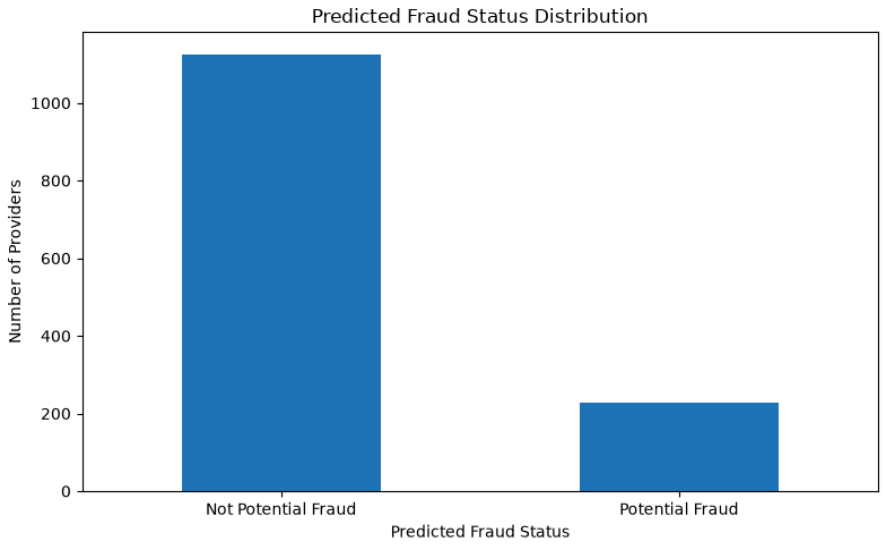
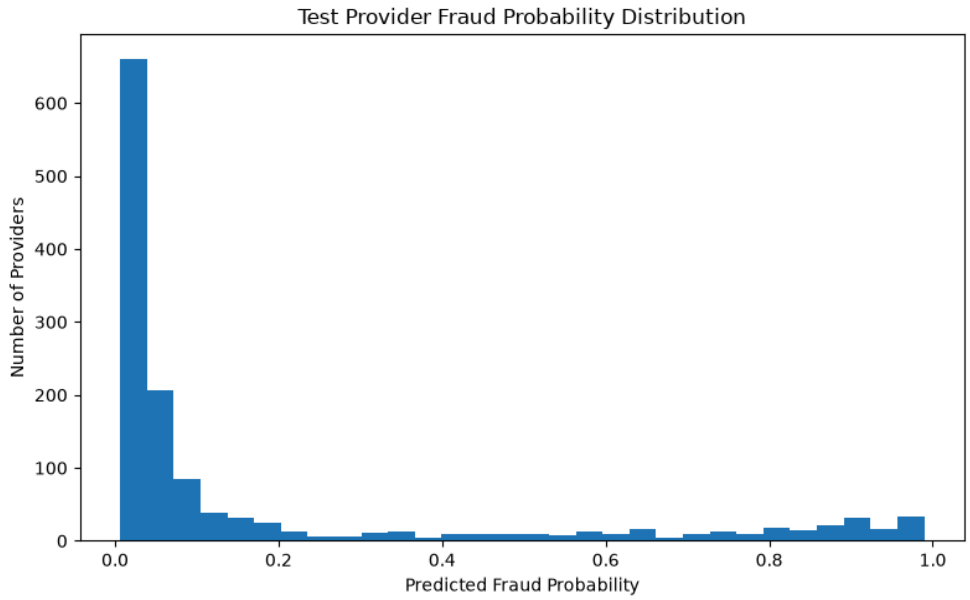


Fig 8. Provider Fraud Probability



11.3 Test Risk-Level Distribution

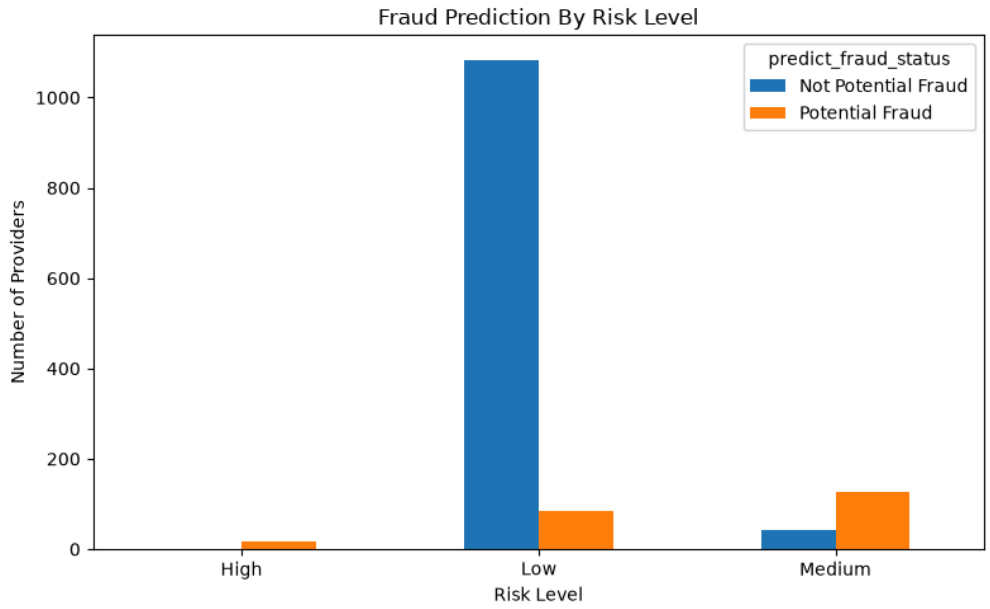
The same seven-feature rule-based scoring framework used for the training data was applied to the test providers using the training-derived thresholds. Providers were classified as Low, Medium, or High risk based on their total risk score.

Among the 1,353 test providers, 1,108 were classified as Low risk, 205 as Medium risk, and 40 as High risk. Of the 40 High-risk providers, 35 were also predicted as Potential Fraud by the Gradient Boosting model. Among Medium-risk providers, 114 were predicted as Potential Fraud, while 78 Potential Fraud predictions occurred within the Low-risk group.

Table 6. Risk Level in Test Data

Risk Level	Not Potential Fraud	Potential Fraud	Total
Low	1,030	78	1,108
Medium	91	114	205
High	5	35	40

Fig 9. Fraud Prediction by Risk Level



11.4 Interpretation of Test Predictions

The test results show substantial agreement between the machine-learning predictions and the rule-based risk framework, particularly within the High-risk group. Most High-risk providers were also classified as Potential Fraud by the Gradient Boosting model, suggesting that providers showing multiple extreme billing and utilization characteristics were more likely to receive elevated model-based fraud predictions.

However, the two methods were not identical. Some providers predicted as Potential Fraud remained in the Low-risk category, indicating that the Gradient Boosting model identified more complex combinations of features that were not captured by the simple threshold-based scoring system. Together, the two approaches provide complementary methods for prioritizing providers for further investigation.

12. Discussion

The findings indicate that billing and utilization behavior were more informative for fraud-risk identification than patient demographics. Potential Fraud providers showed higher claim volumes, reimbursement, inpatient utilization, longer maximum claim durations, and more high-cost claims, while demographic differences were relatively small.

The rule-based framework also showed a clear pattern, with fraud-labeled providers increasing from 2.75% in the Low-risk group to 75.74% in the High-risk group. Among the machine-learning models, Gradient Boosting was selected because it achieved the highest fraud recall (0.90) while maintaining strong F1-score, ROC-AUC, and PR-AUC performance. Feature importance further emphasized total reimbursement, maximum claim duration, and claims per patient as key predictors.

On the unlabeled test dataset, the model predicted 227 of 1,353 providers as Potential Fraud. Of the 40 providers classified as High risk by the rule-based method, 35 were also predicted as Potential Fraud. Together, the two approaches provide complementary tools for prioritizing providers for further review, but the results should be treated as indicators of risk rather than proof of fraud.

13. Limitations

This project has several limitations. First, the analysis was based on a single healthcare fraud dataset, so the findings may not generalize directly to other healthcare systems, populations, or time periods. The target variable represents providers labeled as Potential Fraud, which should be interpreted as a risk indicator rather than confirmed evidence of fraudulent activity. In addition, the training data were highly imbalanced, with far fewer fraud-labeled providers than non-fraud providers.

The feature engineering focused mainly on provider billing, utilization, beneficiary demographics, chronic conditions, claim duration, and high-cost claim behavior. Although the original claims data contained diagnosis and procedure-code information, these variables were not incorporated into the modeling pipeline, which may have limited the ability to detect more detailed clinical billing patterns.

Finally, the separate test dataset did not contain known fraud labels, so the 227 predicted Potential Fraud providers could not be evaluated against actual outcomes. The rule-based risk thresholds were also based on percentile cutoffs from the training data, which may require recalibration when applied to different provider populations.

14. Ethical and Practical Considerations

Healthcare fraud-risk models should be used as screening and prioritization tools rather than as proof of fraudulent behavior. A provider may display unusual billing or utilization patterns for legitimate reasons, such as treating more complex patients or providing specialized services. Therefore, providers identified as high risk should undergo additional review before any administrative or legal action is taken.

The use of demographic and clinical information also requires careful attention to privacy, fairness, and potential bias. Model performance should be monitored to ensure that particular patient or provider groups are not unfairly targeted. In practice, the most appropriate use of the model is to support investigators by identifying providers that may warrant further examination while keeping final decisions under human oversight.

15. Conclusion

This project developed an end-to-end healthcare provider fraud-risk detection framework using claims, beneficiary, and provider data. The analysis combined SQL-based data preparation, Python cleaning, provider-level feature engineering, exploratory analysis, rule-based risk scoring, and machine-learning classification to identify providers with elevated fraud risk.

The results showed that billing and utilization characteristics were the strongest indicators of potential fraud, particularly total reimbursement, claim volume, claims per patient, and maximum claim duration. Among the four models evaluated, Gradient Boosting provided the best overall fraud-screening performance, achieving a fraud recall of 0.90 and identifying 91 of 101 fraud-labeled providers in the validation set.

When applied to the unlabeled test dataset, the model classified 227 of 1,353 providers as Potential Fraud. The rule-based system also identified a smaller group of High-risk providers, most of whom were also flagged by the model. Overall, the project demonstrates how interpretable risk scoring and machine learning can be used together to support provider fraud screening and prioritize cases for further investigation.

References

Gupta, R. A. (n.d.). Healthcare provider fraud detection analysis [Data set]. Kaggle.